

PH 524: Atomic and Molecular Physics

Course Code	PH 524
Course Name	Atomic and Molecular Physics
L-T-P-C	3-0-0-3
Prerequisites	Basic Quantum Mechanics courses
Intended for	Core for M.Sc. and I-Ph.D. in Physics; Elective for others
Mutual Exclusion	None
Approval	9th Senate (May 2016)
Revised	10th Senate (Jun 2016) - revised; 64th BoA (May 2026) - syllabus revised

Preamble

This course introduces the basic ideas of atomic and molecular physics. It teaches students how to apply quantum mechanics and extract information from many-electrons atoms and molecules. The course begins with a review of some of the basic concepts in quantum mechanics and then discusses the time-dependent perturbation theory and its applications. It will then proceed to many-electron atomic systems and then to the concepts associated with molecular spectroscopy.

Course Modules

- **Unit/Topic 1:** Review of perturbation theory, Atoms in electric and magnetic fields (8 hours)
- **Unit/Topic 2:** Time-dependent perturbation theory and application Fermi- Golden rule, Interaction of electromagnetic radiation with single electron atoms, Rabi flopping, Dipole approximation and dipole selection rules, Transition rates, Line broadening mechanisms, spontaneous and stimulated emissions and Einstein coefficients. (14 hours)
- **Unit/Topic 3:** Variational method and applications, two-electron and alkali systems, central field approximation, Slater determinant, Introduction to self-consistent field method, L-S coupling, J-J coupling, Fine structure and Zeeman effect in the alkali spectra, selection rules, Hyperfine interactions. (10 hours)
- **Unit/Topic 4:** Born Oppenheimer Approximation, Rotational and Vibrational and Raman Spectra of molecules-with examples from diatomic molecules. (10 hours)

Laboratory/practical/tutorial Modules: No

Text Books

1. *Physics of atoms and molecules* – Bransden and Joachain (Pearson, second edition, 2011)
2. *A Students' Guide to Atomic Physics* – M. Fox (Cambridge Univ. Press, 2018 First edition)

References

1. *Many-electron theory* – S. Raimes (North-Holland Pub. Co. 1972)
2. *Fundamentals of molecular spectroscopy* – C. Banwell and E. Maccash (Mc Graw Hill, 2013)
3. *Quantum Mechanics* – P. C. Deshmukh (Cambridge University Press 2023)
4. *Quantum Mechanics* – Merzbacher (Third edition, Wiley, 2011)
5. *Quantum mechanics* – Landau and Lifshitz (Butterworth-Heinemann Ltd; 3rd Revised edition (18 December 1981))